What is the best geometry for minimization of MMD?

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1 Introduction

We focus on MMD due to its interpretation as the worse-case integration error in the unit ball of a RKHS.

Notations: For a probability measure π, assume that $\int k(x,x)\,\mathrm{d}\pi(x) < \infty$. We define the associated kernel integral operator $\mathcal{T}_\pi : L_2(\pi) \to L_2(\pi)$ by $(\mathcal{T}_\pi f)(x) = \int k(x,y) f(y)\,\mathrm{d}\pi(y)$ for π-almost every x, where, as usual, elements of $L_2(\pi)$ are understood up to π-almost-everywhere equivalence.

2 Gradient Flows

Gradient flows characterize steepest-descent dynamics on a metric space. Let $(\mu_t)_{t\geq 0}$ be a curve of probability measures and let $\mathcal{F} : \mathcal{P}(\mathbb{R}^d) \to \mathbb{R}$ be an objective functional. The corresponding notion of steepest descent depends on the metric d imposed on $\mathcal{P}(\mathbb{R}^d)$. Consequently, a fixed objective $\mathcal{F}$ generally induces different gradient flows under different choices of d.

We consider the squared MMD objective $\mathcal{F}_{\mathrm{MMD}} = \frac{1}{2}\,\mathrm{MMD}^2(\cdot, \pi)$, motivated by its applications to numerical integration [Chen et al., 2026b] and generative modelling [Li et al., 2017, Arbel et al., 2018, 2019]. We review the gradient flows associated with several standard geometries and discuss their asymptotic convergence in objective value, namely whether $\lim_{t\to\infty} \mathcal{F}_{\mathrm{MMD}}(\mu_t) = 0$. The present section focuses on population-level dynamics, while the next section develops finite-particle approximations.

For later use, the first variation of $\mathcal{F}_{\mathrm{MMD}}$ at μ is $\mathcal{F}'_{\mathrm{MMD}}[\mu](x) = \int k(x,y)\,\mathrm{d}(\mu - \pi)(y)$.

Wasserstein The 2-Wasserstein geometry is defined on $\mathcal{P}_2(\mathbb{R}^d)$, the space of probability measures with finite second moments. For $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$, the 2-Wasserstein distance is $d(\mu,\nu) = W_2(\mu,\nu) := \left(\inf_{\gamma\in\Pi(\mu,\nu)} \int \|x-y\|^2 \mathrm{d}\gamma(x,y)\right)^{\frac{1}{2}}$. Thus, $W_2^2(\mu,\nu)$ is the minimum quadratic cost of transporting μ to ν. Under this geometry, the Wasserstein gradient flow (W-GF) of $\mathcal{F}_{\mathrm{MMD}}$ is [Arbel et al., 2019, Korba et al., 2021]

$$\partial_t \mu_t = \nabla \cdot \left(\mu_t \nabla \mathcal{F}'_{\mathrm{MMD}}[\mu_t]\right). \tag{W-GF}$$

(W-GF) has been widely used for sampling [Chen et al., 2026b] and generative modelling [Galashov et al., 2025], with different kernel choices tailored to the application at hand. Despite its broad empirical success, establishing convergence of (W-GF) remains challenging in general. Several techniques have been proposed to improve its convergence behavior, including noise injection [Arbel et al., 2019, Suzuki et al., 2023, Chen et al., 2026a] and adaptive kernel lengthscales [Kang et al., 2026], but a general convergence theory is still lacking.

Fisher–Rao The Fisher–Rao geometry defines an alternative metric on the space of probability measures. Suppose that $\mu, \nu \in \mathcal{P}(\mathbb{R}^d)$ admit densities with respect to Lebesgue measure, and use the same symbols to denote these densities. Their Fisher–Rao distance is $d(\mu, \nu) = d_{\mathrm{FR}}(\mu, \nu) := 2 \arccos \left(\int \sqrt{\mu(x)\nu(x)} \, \mathrm{d}x \right)$. The corresponding Fisher–Rao gradient flow (FR-GF) of $\mathcal{F}_{\mathrm{MMD}}$ is

$$\partial_t \mu_t = -\mu_t \left(\mathcal{F}'_{\mathrm{MMD}}[\mu_t] - \int \mathcal{F}'_{\mathrm{MMD}}[\mu_t] \mathrm{d}\mu_t \right). \tag{FR-GF}$$

Wasserstein–Fisher–Rao The W-GF in (W-GF) transports mass through space, whereas the FR-GF in (FR-GF) redistributes mass locally without spatial transport. The Wasserstein–Fisher–Rao gradient flow (WFR-GF) combines these two mechanisms. For $\alpha, \beta > 0$, the resulting dynamics are

$$\partial_t \mu_t = \alpha \nabla \cdot \left(\mu_t \nabla \mathcal{F}'_{\mathrm{MMD}}[\mu_t] \right) - \beta \mu_t \left(\mathcal{F}'_{\mathrm{MMD}}[\mu_t] - \int \mathcal{F}'_{\mathrm{MMD}}[\mu_t] \mathrm{d}\mu_t \right). \tag{WFR-GF}$$

Interaction Force Transport Recent work has considered the geometry induced by the MMD distance, $d(\mu, \nu) = \mathrm{MMD}(\mu, \nu)$ [Zhu, 2024, Zhu and Mielke, 2024, Gladin et al., 2024]. For a kernel k and a probability measure μ, let $\mathcal{T}_\mu : L^2(\mu) \to L^2(\mu)$ denote the associated kernel integral operator. The first variation of the MMD objective can be written as $\mathcal{F}'_{\mathrm{MMD}} = \int k(\cdot, y) \, \mathrm{d}(\mu - \pi)(y) = \mathcal{T}_\mu(1 - \mathrm{d}\mu/\mathrm{d}\pi)$. The resulting Interaction-Force Transport (IFT) gradient flow is [Gladin et al., 2024]

$$\partial_t \mu_t = -\mu_t \mathcal{T}_{\mu_t}^{-1} \mathcal{F}'_{\mathrm{MMD}}[\mu_t] = -(\mu_t - \pi). \tag{IFT-GF}$$

This representation relies on the cancellation of $\mathcal{T}_{\mu_t}^{-1}$ with $\mathcal{T}_{\mu_t}$ in $\mathcal{F}'_{\mathrm{MMD}}$. The present work is restricted to $\mathcal{F}_{\mathrm{MMD}}$; extending this construction to more general objectives $\mathcal{F}$ is beyond its scope. The IFT-GF redistributes mass without spatial movement.

Wasserstein Interaction Force Transport Combining W-GF and IFT-GF, analogously to (WFR-GF), yields

$$\partial_t \mu_t = \alpha \nabla \cdot \left(\mu_t \nabla \mathcal{F}'[\mu_t] \right) - \beta \mathcal{T}^{-1} \mathcal{F}'[\mu_t] \tag{WIFT-GF}$$

Both IFT-GF and WIFT-GF converge exponentially in squared MMD [Gladin et al., 2024]; specifically, $\mathrm{MMD}^2(\mu_t, \pi) \le e^{-2\beta t} \, \mathrm{MMD}^2(\mu_0, \pi)$. Notably, this convergence holds without conditions on the target distribution π. The closest comparable result in Tian et al. [2026] requires additional regularization.

3 Particle Descent

If $\mathcal{F}$ is the MMD distance $\mathcal{F}(\cdot) = \mathrm{MMD}^2(\cdot, \pi)$, then the first variation $\mathcal{F}'[\mu](\cdot) = \int k(x, \cdot) \mathrm{d}(\mu - \pi) = \mathcal{T}(\mu - \pi)$. Then, we can derive the particle descent scheme of the corresponding gradient flows.

Let $\mu_{N,t} = \sum_{i=1}^N w_t^{(i)} \delta_{x_t^{(i)}}$ be a weighted empirical measure at iteration t. Write $\boldsymbol{X}_t = (x_t^{(1)}, \ldots, x_t^{(N)})$ and $\boldsymbol{w}_t = (w_t^{(1)}, \ldots, w_t^{(N)})$. Denote $m_\pi = \int k(x, \cdot) \mathrm{d}\pi$ as the kernel mean embedding. Denote the kernel-mean residual and its vectorized counterpart as,

$$h_{\boldsymbol{X},\boldsymbol{w}}(x) := \sum_{j=1}^N w^{(j)} k(x^{(j)}, x) - m_\pi(x) \in \mathbb{R}, \quad h_{\boldsymbol{X},\boldsymbol{w}}(\boldsymbol{v}) = [h_{\boldsymbol{X},\boldsymbol{w}}(v^{(1)}), \ldots, h_{\boldsymbol{X},\boldsymbol{w}}(v^{(N)})]^\top \in \mathbb{R}^N.$$

Define

$$E(\boldsymbol{X}, \boldsymbol{w}) = \frac{1}{2} \operatorname{MMD}^2 \left(\sum_{i=1}^{N} w^{(i)} \delta_{x^{(i)}}, \pi \right).$$

Euclidean geometry The most straightforward approach to optimizing the particle locations and weights is to apply gradient descent jointly to $\boldsymbol{X}$ and $\boldsymbol{w}$. For consistency with the update schemes introduced later, we adopt a Gauss–Seidel-type block gradient descent scheme, in which the particle locations are updated first and the weights are subsequently updated using the new locations $\boldsymbol{X}_{t+1}$. In particular, let $\tau > 0$ be the step size. We consider $\boldsymbol{X}_{t+1} = \boldsymbol{X}_t - \tau \nabla_{\boldsymbol{X}} E(\boldsymbol{X}_t, \boldsymbol{w}_t)$ and $\boldsymbol{w}_{t+1} = \boldsymbol{w}_t - \tau \nabla_{\boldsymbol{w}} E(\boldsymbol{X}_t, \boldsymbol{w}_t)$. Expanding the gradients yields the following explicit update scheme: for each $i \in [N]$,

$$x_{t+1}^{(i)} = x_t^{(i)} - \tau w_t^{(i)} \nabla h_{\boldsymbol{X}_t, \boldsymbol{w}_t} \left(x_t^{(i)} \right), \quad w_{t+1}^{(i)} = w_t^{(i)} - \tau h_{\boldsymbol{X}_{t+1}, \boldsymbol{w}_t} \left(x_{t+1}^{(i)} \right) \tag{1}$$

Wasserstein Fisher-Rao geometry

$$x_{t+1}^{(i)} = x_t^{(i)} - \tau \alpha \nabla h_{\boldsymbol{X}_t, \boldsymbol{w}_t} \left(x_t^{(i)} \right), \quad w_{t+1}^{(i)} = w_t^{(i)} - \tau \beta w_t^{(i)} h_{\boldsymbol{X}_{t+1}, \boldsymbol{w}_t} \left(x_{t+1}^{(i)} \right). \tag{2}$$

Wasserstein IFT geometry The locations are first updated as the same as above, however, the weight update scheme require knowledge of the true density π. In reality, the weights are updated instead via an implicit minimizing scheme.

$$x_{t+1}^{(i)} = x_t^{(i)} - \tau \alpha \nabla h_{\boldsymbol{X}_t, \boldsymbol{w}_t} \left(x_t^{(i)} \right), \tag{3}$$

$$\boldsymbol{w}_{t+1} = \arg \min_{\boldsymbol{w} \in \Delta^N} \left\{ \frac{1}{2} \operatorname{MMD}^2 \left(\sum_{i=1}^{N} w^{(i)} \delta_{x_{t+1}^{(i)}}, \pi \right) + \frac{1}{2(\beta\tau)} \operatorname{MMD}^2 \left(\sum_{i=1}^{N} w^{(i)} \delta_{x_{t+1}^{(i)}}, \mu_{N,t} \right) \right\}. \tag{4}$$

Interestingly, letting $K_{t+1} = K(\boldsymbol{X}_{t+1}, \boldsymbol{X}_{t+1}) \in \mathbb{R}^{N \times N}$ be the Gram matrix, the weight problem is

$$\min_{\boldsymbol{w} \in \Delta^N} \left\{ \frac{1}{2} \boldsymbol{w}^\top K_{t+1} \boldsymbol{w} - m_\pi(\boldsymbol{X}_{t+1})^\top \boldsymbol{w} + \frac{1}{2\eta} \left(\boldsymbol{w} - \boldsymbol{w}_t \right)^\top K_{t+1} \left(\boldsymbol{w} - \boldsymbol{w}_t \right) \right\},$$

By definition of kernel quadrature, there is $m_\pi(\boldsymbol{X}_{t+1}) = K_{t+1} \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})$. And the solution is, when ignoring the simplex constraint, $\boldsymbol{w}_{t+1} = \frac{1}{1+\eta} \boldsymbol{w}_t + \frac{\eta}{1+\eta} \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})$ which can be equivalently written as,

$$\boldsymbol{w}_{t+1} = \boldsymbol{w}_t - \frac{1}{1 + (\tau\beta)} {K_{t+1}}^{-1} \left(h_{\boldsymbol{X}_{t+1}, \boldsymbol{w}_t} \left(\boldsymbol{X}_{t+1} \right) \right).$$

To state the particle updates compactly, Table 1 gives the location and weight updates explicitly for the three geometries, direct Euclidean gradient descent, and MSIP [Belhadji et al., 2026]. In the WFR scheme, the transport step precedes the weight step, which is evaluated at the updated particle locations. The WFR weight update is obtained by applying forward Euler discretization to the Fisher–Rao reaction term in Eq. (WFR-GF).

The Wasserstein row uses uniform initial weights, which remain equal to $1/N$ under transport; arbitrary initial weights would likewise remain fixed. The centering term in the WFR update preserves $\sum_{i=1}^{N} w_t^{(i)} = 1$, while nonnegativity additionally requires $\tau\beta$ to be sufficiently small. The MMD row is the WIFT particle implementation used here: it combines the Wasserstein location

Table 1: Location and weight updates for MMD minimization under Wasserstein, WFR, and MMD geometries, together with direct Euclidean gradient descent and MSIP.

Geometry or method	Location and weight updates
Wasserstein	$x_{t+1}^{(i)} = x_t^{(i)} - \tau\alpha\nabla h_{\boldsymbol{X}_t,\boldsymbol{w}_t}\left(x_t^{(i)}\right)$, $w_{t+1}^{(i)} = w_t^{(i)} = \frac{1}{N}$.
WFR	$x_{t+1}^{(i)} = x_t^{(i)} - \tau\alpha\nabla h_{\boldsymbol{X}_t,\boldsymbol{w}_t}\left(x_t^{(i)}\right)$, $w_{t+1}^{(i)} = w_t^{(i)} - \tau\beta w_t^{(i)}\left[h_{\boldsymbol{X}_{t+1},\boldsymbol{w}_t}\left(x_{t+1}^{(i)}\right) - \sum_{j=1}^{N} w_t^{(j)} h_{\boldsymbol{X}_{t+1},\boldsymbol{w}_t}\left(x_{t+1}^{(j)}\right)\right]$.
MMD (IFT)	$x_{t+1}^{(i)} = x_t^{(i)} - \tau\alpha\nabla h_{\boldsymbol{X}_t,\boldsymbol{w}_t}\left(x_t^{(i)}\right)$, $\boldsymbol{w}_{t+1} = \frac{1}{1+\eta}\boldsymbol{w}_t + \frac{\eta}{1+\eta}\boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})$.
Direct GD	$x_{t+1}^{(i)} = x_t^{(i)} - \tau\alpha w_t^{(i)}\nabla h_{\boldsymbol{X}_t,\boldsymbol{w}_t}\left(x_t^{(i)}\right)$, $w_{t+1}^{(i)} = w_t^{(i)} - \tau\beta h_{\boldsymbol{X}_t,\boldsymbol{w}_t}\left(x_t^{(i)}\right)$.
MSIP	For a squared-exponential kernel, $x_{t+1}^{(i)} = (1-\tau)x_t^{(i)} + \frac{\tau}{w_{\mathrm{KQ}}^{(i)}(\boldsymbol{X}_t)}\sum_{j=1}^{N}\left[K(\boldsymbol{X}_t)^{-1}\right]_{ij}\int xk\left(x, x_t^{(j)}\right)\mathrm{d}\pi(x)$, $\boldsymbol{w}_{t+1} = \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})$.

update with the IFT weight update in Eqs. (**??**)–(**??**). Its displayed closed form is obtained when the simplex constraint in Eq. (**??**) is ignored. The direct-gradient row performs simultaneous Euclidean descent in both $\boldsymbol{X}$ and $\boldsymbol{w}$; its location update differs from the Wasserstein update by the factor $w_t^{(i)}$. This weight update is unconstrained and must be projected if the iterates are required to remain in Δ^N. In contrast, MSIP is a damped fixed-point method for the stationary conditions of the unbalanced WFR flow, rather than a time discretization of the normalized WFR flow in Eq. (WFR-GF). Its update requires $K(\boldsymbol{X}_t)$ to be invertible and the kernel quadrature weights to be positive.

Definition 3.1 (Weighted Euclidean Geometry)**.** *Let $m \in \mathbb{R}$ and let $G \in \mathbb{R}^{m\times m}$ be a symmetric positive definite matrix. We equip the space $\mathbb{R}^m$ with the G-weighted Euclidean inner product $\langle u, v\rangle_G := u^\top G v$ and with the induced norm $\|u\|_G^2 := u^\top G u$. For a differentiable function $f : \mathbb{R}^m \to \mathbb{R}$, its gradient with respect to the G-weighted Euclidean geometry, denoted by $\mathrm{grad}_G f$, is defined as: for any $u \in \mathbb{R}^m$,*

$$\frac{\mathrm{d}}{\mathrm{d}t} f(x+tu)\bigg|_{t=0} = \langle \mathrm{grad}_G f(x), u\rangle_G .$$

Equivalently, $\mathrm{grad}_G f(x) = G^{-1}\nabla f(x)$ where ∇f denotes the standard Euclidean gradient.

Proposition 3.2 (Smoothness under Euclidean and weighted Euclidean geometries)**.** *Let $\boldsymbol{w} \in \Delta^N$ with $w^{(i)} > 0$ for every i and $\sum_{i=1}^N w^{(i)} = 1$. Suppose that k is symmetric and twice continuously differentiable, with $\sup_{x,y\in\mathbb{R}^d}\|\nabla_{11}^2 k(x,y)\|_{\mathrm{op}} = C_1 < \infty$ and $\sup_{x,y\in\mathbb{R}^d}\|\nabla_{12}^2 k(x,y)\|_{\mathrm{op}} = C_2 < \infty$.*

Define

$$E(\boldsymbol{X}, \boldsymbol{w}) = \frac{1}{2}\operatorname{MMD}^2\left(\sum_{i=1}^N w^{(i)}\delta_{x^{(i)}}, \pi\right).$$

Then $E(\cdot, \boldsymbol{w})$ is $\max_{1\le i\le N} w^{(i)}(2C_1 + C_2)$-smooth under the standard Euclidean geometry; and is $(2C_1{+}C_2)$-smooth under the $\boldsymbol{w}$-weighted Euclidean geometry. Moreover, $E(\boldsymbol{X}, \cdot)$ is $\lambda_{\max}(K(\boldsymbol{X}, \boldsymbol{X}))$-smooth under the standard Euclidean geometry; and is 1-smooth under the $K(\boldsymbol{X}, \boldsymbol{X})$-weighted Euclidean geometry.

From the above proposition, we see that the right smoothness notion to consider is the weighted Euclidean geometry instead of the standard Euclidean geometry.

The direct KQ update instantaneously minimizes the MMD over the weights at each particle configuration, whereas the IFT update is a proximal tracking scheme that interpolates between the previous weights and the instantaneous KQ solution. The latter sacrifices instantaneous optimality in exchange for a controlled and stable evolution of the weights, consistent with the underlying continuous-time IFT dynamics.

4 Convergence

Theorem 4.1 (Monotone improvement)**.** *At iteration $t \in \mathbb{N}$, let $\mu_{N,t} = \sum_{i=1}^N w_t^{(i)}\delta_{x_t^{(i)}}$ with $\boldsymbol{X}_t = (x_t^{(1)}, \dots, x_t^{(N)})$, and $\boldsymbol{w}_t = (w_t^{(1)}, \dots, w_t^{(N)})$.*

Assume that k is twice continuously differentiable, that $K(\boldsymbol{X}_t)$ is positive definite for every t, and that the weights remain uniformly positive, namely $w_t^{(i)} \ge \underline{w} > 0$ for every i, t. Assume moreover that, for every admissible $\boldsymbol{w}$, the map $\boldsymbol{X} \mapsto \mathcal{E}(\boldsymbol{X}, \boldsymbol{w})$ has an L_X-Lipschitz gradient on a set containing all iterates. If $\tau\alpha \le 2\underline{w}/L_X$, then for every $t \ge 0$,

$$\operatorname{MMD}\left(\mu_{N,t+1}, \pi\right) \le \operatorname{MMD}\left(\mu_{N,t}, \pi\right). \tag{5}$$

Lemma 4.2 (Uniform control of the kernel quadrature residual)**.** *Suppose that the assumptions of Theorem 4.1 hold and that the initial weights are chosen as $\boldsymbol{w}_0 = \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_0)$. Then, for every $t \ge 0$, $\operatorname{MMD}\left(\sum_{i=1}^N w_{\mathrm{KQ}}^{(i)}(\boldsymbol{X}_t)\delta_{x_t^{(i)}}, \pi\right) \le \operatorname{MMD}\left(\sum_{i=1}^N w_{\mathrm{KQ}}^{(i)}(\boldsymbol{X}_0)\delta_{x_0^{(i)}}, \pi\right)$.*

Theorem 4.3 (Convergence to the kernel quadrature error)**.** *Suppose that the assumptions of Theorem 4.1 hold, and that the initial weights are chosen as $\boldsymbol{w}_0 = \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_0)$. Then, the WIFT iterates satisfy*

$$\operatorname{MMD}^2(\mu_{N,t}, \pi) \le \frac{1}{(1+\eta)^{2t}}\operatorname{MMD}^2(\mu_{N,0}, \pi) + \left(1 - \frac{1}{(1+\eta)^{2t}}\right)\operatorname{MMD}^2\left(\sum_{i=1}^N w_{\mathrm{KQ}}^{(i)}(\boldsymbol{X}_0)\delta_{x_0^{(i)}}, \pi\right).$$

Remark 4.4. *Theorem 4.3 separates the optimization error from the statistical error of the initial kernel quadrature rule. In particular, if $\operatorname{MMD}(\sum_{i=1}^N w_{\mathrm{KQ}}^{(i)}(\boldsymbol{X}_0)\delta_{x_0^{(i)}}, \pi) = O_{\mathbb{P}}(N^{-s/d})$, then*

$$\operatorname{MMD}(\mu_{N,t}, \pi) \le \frac{1}{(1+\eta)^t}\operatorname{MMD}(\mu_{N,0}, \pi) + O_{\mathbb{P}}(N^{-s/d}).$$

Thus, the optimization error decays geometrically until it reaches the kernel quadrature error floor. If, in addition, $\operatorname{MMD}(\mu_{N,0}, \pi) = O_{\mathbb{P}}(1)$, then choosing $t \ge \frac{s}{d\log(1+\eta)}\log N$ is sufficient to obtain $\operatorname{MMD}(\mu_{N,t}, \pi) = O_{\mathbb{P}}(N^{-s/d})$.

5 Proofs

Proof of Proposition 3.2. Recall that

$$E(\boldsymbol{X}, \boldsymbol{w}) = \frac{1}{2} \sum_{i,j=1}^{N} w^{(i)} w^{(j)} k(x^{(i)}, x^{(j)}) - \sum_{i=1}^{N} w^{(i)} m_\pi(x^{(i)}) + \text{const.}$$

For any $\boldsymbol{u} = (u^{(1)}, \dots, u^{(N)}) \in (\mathbb{R}^d)^N$, direct differentiation gives

$$\begin{aligned}\nabla^2_{\boldsymbol{X}} E(\boldsymbol{X}, \boldsymbol{w})[\boldsymbol{u}, \boldsymbol{u}] = & \sum_{i,j=1}^{N} w^{(i)} w^{(j)} (u^{(i)})^\top \nabla^2_{11} k(x^{(i)}, x^{(j)}) u^{(i)} \\ & + \sum_{i,j=1}^{N} w^{(i)} w^{(j)} (u^{(i)})^\top \nabla^2_{12} k(x^{(i)}, x^{(j)}) u^{(j)} - \sum_{i=1}^{N} w^{(i)} (u^{(i)})^\top \nabla^2 m_\pi(x^{(i)}) u^{(i)}.\end{aligned}$$

Since $m_\pi(x) = \int k(x,y)\, \mathrm{d}\pi(y)$ and π is a probability measure, we have

$$\|\nabla^2 m_\pi(x)\|_{\text{op}} = \left\| \int \nabla^2_{11} k(x,y)\, \mathrm{d}\pi(y) \right\|_{\text{op}} \le \sup_{x,y \in \mathbb{R}^d} \|\nabla^2_{11} k(x,y)\|_{\text{op}} = C_1.$$

Using $\sum_{j=1}^{N} w^{(j)} = 1$, it follows that

$$\left| \sum_{i,j=1}^{N} w^{(i)} w^{(j)} (u^{(i)})^\top \nabla^2_{11} k(x^{(i)}, x^{(j)}) u^{(i)} - \sum_{i=1}^{N} w^{(i)} (u^{(i)})^\top \nabla^2 m_\pi(x^{(i)}) u^{(i)} \right| \le 2C_1 \sum_{i=1}^{N} w^{(i)} \|u^{(i)}\|^2.$$

For the mixed term,

$$\begin{aligned}\left| \sum_{i,j=1}^{N} w^{(i)} w^{(j)} (u^{(i)})^\top \nabla^2_{12} k(x^{(i)}, x^{(j)}) u^{(j)} \right| &\le \sup_{x,y \in \mathbb{R}^d} \|\nabla^2_{12} k(x,y)\|_{\text{op}} \left(\sum_{i=1}^{N} w^{(i)} \|u^{(i)}\| \right)^2 \\ &= C_2 \left(\sum_{i=1}^{N} w^{(i)} \|u^{(i)}\| \right)^2.\end{aligned}$$

By Cauchy–Schwarz and $\sum_{i=1}^{N} w^{(i)} = 1$, we have $\left(\sum_{i=1}^{N} w^{(i)} \|u^{(i)}\|\right)^2 \le \sum_{i=1}^{N} w^{(i)} \|u^{(i)}\|^2$. Therefore,

$$\left|\nabla^2_{\boldsymbol{X}} E(\boldsymbol{X}, \boldsymbol{w})[\boldsymbol{u}, \boldsymbol{u}]\right| \le (2C_1 + C_2) \sum_{i=1}^{N} w^{(i)} \|u^{(i)}\|^2 = (2C_1 + C_2)\, \|\boldsymbol{u}\|^2_{\boldsymbol{w}}.$$

The last step holds by the definition of $\boldsymbol{w}$-weighted Euclidean geometry in Definition 3.1. This proves smoothness under the weighted Euclidean geometry.

For the standard Euclidean geometry, since $\sum_{i=1}^{N} w^{(i)} \|u^{(i)}\|^2 \le \left(\max_{1 \le i \le N} w^{(i)}\right) \sum_{i=1}^{N} \|u^{(i)}\|^2 = \left(\max_{1 \le i \le N} w^{(i)}\right) \|\boldsymbol{u}\|_2^2$. The proof is concluded. □

Proof of Theorem 4.1. For fixed $\boldsymbol{X}$ and $\boldsymbol{w}$, let $g^{(i)}(\boldsymbol{X}, \boldsymbol{w}) = \sum_{j=1}^{N} w^{(j)} \nabla_2 k(x^{(j)}, x^{(i)}) - \nabla m_\pi(x^{(i)})$. Define the MMD energy by

$$\mathcal{E}(\boldsymbol{X}, \boldsymbol{w}) = \frac{1}{2} \mathrm{MMD}^2 \left(\sum_{i=1}^{N} w^{(i)} \delta_{x^{(i)}}, \pi \right), \tag{6}$$

Differentiating the MMD energy with respect to $x^{(i)}$ and using symmetry of the kernel gives $\nabla_{x^{(i)}}\mathcal{E}(\boldsymbol{X},\boldsymbol{w}) = w^{(i)}g^{(i)}(\boldsymbol{X},\boldsymbol{w})$. Hence, the position update can be written as $x_{t+1}^{(i)} = x_t^{(i)} - \frac{\tau\alpha}{w_t^{(i)}}\nabla_{x^{(i)}}\mathcal{E}(\boldsymbol{X}_t,\boldsymbol{w}_t)$. Thus, the Wasserstein substep is a diagonally preconditioned gradient step for $\boldsymbol{X} \mapsto \mathcal{E}(\boldsymbol{X},\boldsymbol{w}_t)$. By the L_X-smoothness of $\mathcal{E}(\cdot,\boldsymbol{w}_t)$, the descent lemma gives

$$\begin{aligned}
\mathcal{E}(\boldsymbol{X}_{t+1},\boldsymbol{w}_t) &\leq \mathcal{E}(\boldsymbol{X}_t,\boldsymbol{w}_t) + \sum_{i=1}^N \left\langle \nabla_{x^{(i)}}\mathcal{E}(\boldsymbol{X}_t,\boldsymbol{w}_t), x_{t+1}^{(i)} - x_t^{(i)} \right\rangle + \frac{L_X}{2}\sum_{i=1}^N \left\| x_{t+1}^{(i)} - x_t^{(i)} \right\|^2 \\
&= \mathcal{E}(\boldsymbol{X}_t,\boldsymbol{w}_t) - \tau\alpha \sum_{i=1}^N \left(w_t^{(i)} - \frac{L_X\tau\alpha}{2} \right) \left\| g^{(i)}(\boldsymbol{X}_t,\boldsymbol{w}_t) \right\|^2 .
\end{aligned} \tag{7}$$

Since $w_t^{(i)} \geq \underline{w}$ and $\tau\alpha \leq 2\underline{w}/L_X$, we have $w_t^{(i)} - L_X\tau\alpha/2 \geq 0$. Therefore, $\mathcal{E}(\boldsymbol{X}_{t+1},\boldsymbol{w}_t) \leq \mathcal{E}(\boldsymbol{X}_t,\boldsymbol{w}_t)$. This proves that the particle update in Eq. (**??**) is decreasing the MMD value.

We next consider the weight update. For fixed $\boldsymbol{X}$, the dependence of the MMD energy on $\boldsymbol{w}$ is quadratic: $\mathcal{E}(\boldsymbol{X},\boldsymbol{w}) = \frac{1}{2}\boldsymbol{w}^\top K(\boldsymbol{X})\boldsymbol{w} - m_\pi(\boldsymbol{X})^\top\boldsymbol{w}$+const. Since $K(\boldsymbol{X})$ is positive definite, the unique minimizer is $\boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}) = K(\boldsymbol{X})^{-1}m_\pi(\boldsymbol{X})$. Completing the square gives

$$\mathcal{E}(\boldsymbol{X},\boldsymbol{w}) = \mathcal{E}\left(\boldsymbol{X},\boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X})\right) + \frac{1}{2}\left(\boldsymbol{w} - \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X})\right)^\top K(\boldsymbol{X})\left(\boldsymbol{w} - \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X})\right). \tag{8}$$

By the weight update,

$$\boldsymbol{w}_{t+1} - \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1}) = \frac{1}{1+\eta}\left(\boldsymbol{w}_t - \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})\right). \tag{9}$$

On the other hand, for fixed particle locations $\boldsymbol{X}_{t+1}$, the KQ decomposition gives

$$\begin{aligned}
&\mathcal{E}(\boldsymbol{X}_{t+1},\boldsymbol{w}_{t+1}) - \mathcal{E}\left(\boldsymbol{X}_{t+1},\boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})\right) \\
&\quad = \frac{1}{2}\left(\boldsymbol{w}_{t+1} - \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})\right)^\top K_{t+1}\left(\boldsymbol{w}_{t+1} - \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})\right).
\end{aligned} \tag{10}$$

Substituting (9) into (10) yields

$$\begin{aligned}
&\mathcal{E}(\boldsymbol{X}_{t+1},\boldsymbol{w}_{t+1}) - \mathcal{E}\left(\boldsymbol{X}_{t+1},\boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})\right) \\
&\quad = \frac{1}{2(1+\eta)^2}\left(\boldsymbol{w}_t - \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})\right)^\top K_{t+1}\left(\boldsymbol{w}_t - \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})\right) \\
&\quad = \frac{1}{(1+\eta)^2}\left[\mathcal{E}(\boldsymbol{X}_{t+1},\boldsymbol{w}_t) - \mathcal{E}\left(\boldsymbol{X}_{t+1},\boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})\right)\right].
\end{aligned} \tag{11}$$

Since $(1+\eta)^{-2} \leq 1$ and $\boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})$ minimizes $\boldsymbol{w} \mapsto \mathcal{E}(\boldsymbol{X}_{t+1},\boldsymbol{w})$, it follows that $\mathcal{E}(\boldsymbol{X}_{t+1},\boldsymbol{w}_{t+1}) \leq \mathcal{E}(\boldsymbol{X}_{t+1},\boldsymbol{w}_t)$. Combining the two substeps yields

$$\mathcal{E}(\boldsymbol{X}_{t+1},\boldsymbol{w}_{t+1}) \leq \mathcal{E}(\boldsymbol{X}_{t+1},\boldsymbol{w}_t) \leq \mathcal{E}(\boldsymbol{X}_t,\boldsymbol{w}_t). \tag{12}$$

Hence, $\mathcal{E}(\boldsymbol{X}_t,\boldsymbol{w}_t)$ is nonincreasing, and the proof is concluded. □

Proof of Lemma 4.2. Recall the definition that $\mathcal{E}(\boldsymbol{X},\boldsymbol{w}) = \frac{1}{2}\,\mathrm{MMD}^2\left(\sum_{i=1}^N w^{(i)}\delta_{x^{(i)}},\pi\right)$. For any fixed particle locations $\boldsymbol{X}_t$, the kernel quadrature weights $\boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_t)$ minimize the MMD energy over the weights. Hence, $\mathcal{E}\left(\boldsymbol{X}_t,\boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_t)\right) \leq \mathcal{E}\left(\boldsymbol{X}_t,\boldsymbol{w}_t\right)$. On the other hand, by the monotonicity of the WIFT iteration established in Theorem 4.1, we have $\mathcal{E}\left(\boldsymbol{X}_t,\boldsymbol{w}_t\right) \leq \mathcal{E}\left(\boldsymbol{X}_0,\boldsymbol{w}_0\right)$. Since $\boldsymbol{w}_0 = \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_0)$, we have $\mathcal{E}(\boldsymbol{X}_0,\boldsymbol{w}_0) = \mathcal{E}(\boldsymbol{X}_0,\boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_0))$. Combining the previous inequalities gives

$$\mathcal{E}\left(\boldsymbol{X}_t,\boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_t)\right) \leq \mathcal{E}\left(\boldsymbol{X}_t,\boldsymbol{w}_t\right) \leq \mathcal{E}\left(\boldsymbol{X}_0,\boldsymbol{w}_0\right) = \mathcal{E}\left(\boldsymbol{X}_0,\boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_0)\right). \tag{13}$$

The proof is thus concluded. □

Proof of Theorem 4.3. From the exact contraction of the weight update in Eq. (11), together with descent of the Wasserstein position update, we have

$$\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}_{t+1}) \leq \frac{1}{(1+\eta)^2}\mathcal{E}(\boldsymbol{X}_t, \boldsymbol{w}_t) + \left(1 - \frac{1}{(1+\eta)^2}\right)\mathcal{E}\left(\boldsymbol{X}_{t+1}, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})\right). \tag{14}$$

By Lemma 4.2, $\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_{t+1})) \leq \mathcal{E}(\boldsymbol{X}_0, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_0))$, and therefore

$$\mathcal{E}(\boldsymbol{X}_{t+1}, \boldsymbol{w}_{t+1}) \leq \frac{1}{(1+\eta)^2}\mathcal{E}(\boldsymbol{X}_t, \boldsymbol{w}_t) + \left(1 - \frac{1}{(1+\eta)^2}\right)\mathcal{E}\left(\boldsymbol{X}_0, \boldsymbol{w}_{\mathrm{KQ}}(\boldsymbol{X}_0)\right). \tag{15}$$

The proof is concluded. □

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